

Spectral Characterization of QMA

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Abstract

We explore the intermediate region between BQP and QMA-complete sets using the concept of limited quantum non-determinism and propose an infinite hierarchy of quantum complexity classes between them. Each level in the hierarchy is defined by the size of the quantum circuit required to decide a problem at that level. We also provide an equivalent definition of the hierarchy in the quantum Turing model. Furthermore, we prove that the hierarchy is robust in a certain relativized setting.

Additionally, we propose a complete (promise) problem for each level in the infinite hierarchy named parametrized Guided Local Hamiltonian (GLH_i). Subsequently, we define and quantify a Fourier spectral concentration of the GLH_i problem. Ultimately, this value is used to design a quantum algorithm that harnesses it. To design the quantum algorithm, we repurpose the quantum phase estimation algorithm and incorporate suitable classical pre- and post-processing techniques. As expected, the algorithm's execution time worsens as the spectral concentration of a given problem decreases.

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1 The Internal Structure of QMA

We characterize the complexity of natural problems in QMA through a hierarchy starting from BQP and extending to problems complete for QMA. We start with a quantum circuit-based definition of our hierarchy, building on the concept of promise problems. We will now review some basic definitions.

1.1 Preliminary definitions

We will mostly deal with a promise problem, which is a generalization of the notion of a language.

Definition 1.1 (Promise problem). *A promise problem $\mathcal{A} \equiv \{\mathcal{A}_{yes}, \mathcal{A}_{no}\}$ where $\mathcal{A}_{yes}$ and $\mathcal{A}_{no}$ are a set of accepted and a set of rejected inputs, respectively. Additionally, the sets $\mathcal{A}_{yes}, \mathcal{A}_{no} \subseteq \{0, 1\}^*$. A set of invalid inputs are defined as $\mathcal{A}_{inv} \equiv \{0, 1\}^* \setminus \{\mathcal{A}_{yes} \cup \mathcal{A}_{no}\}$ which may not be an empty set. A promise problem becomes a language as a special case when $\mathcal{A}_{inv}$ is the empty set.*

It should be noted that an algorithm deciding a promise problem is usually not provided with an input in $x \in \mathcal{A}_{inv}$. Nevertheless, if invalid inputs are provided to the algorithm, it may accept or reject them arbitrarily.

Since a quantum circuit of fixed size can only take an input of fixed size, a promise problem is decided by a family of circuits that consists of a set of circuits $\{Q_n\}_{n \in \mathbb{N}}$, each for a different input size. It is formally defined as follows.

Definition 1.2 ($U(n)$ -uniform quantum circuit family). *Let $U(n) : \mathbb{N} \rightarrow \mathbb{N}$ be a function. A family of quantum circuits $\{Q_n\}_{n \in \mathbb{N}}$ is called $U(n)$ -uniform if there exists a classical Turing machine that takes 1^n as input and outputs a classical description of the quantum circuit Q_n of size $O(U(n))$ in time at most $\text{poly}(U(n))$.*

When the function $U(n)$ is some polynomial, this special case is called a P -uniform quantum circuit. Building upon these definitions, we review the quantum circuit-based definition of a set of promise problems decided by a uniformly generated quantum circuit of a certain size.

Definition 1.3 ($\text{BQSIZE}(s(n))$). *A promise problem $\mathcal{A} \equiv \{\mathcal{A}_{yes}, \mathcal{A}_{no}\}$ is in $\text{BQSIZE}(s(n))$ if and only if there exists a $\text{poly}(s(n))$ -uniform family of quantum circuits $\{Q_n\}_{n \in \mathbb{N}}$ such that if $x \in \mathcal{A}_{yes}$, then*

$$\Pr[Q_n \text{ accepts } x] \geq 2/3,$$

while, if $x \notin \mathcal{A}_{no}$, then

$$\Pr[M \text{ accepts } x] \leq 1/3.$$

$\text{BQSIZE}[s(n)]$ is defined as the class of all promise problems decided by quantum circuits of size $O(s(n))$.

This definition is an adapted version of a definition for $\text{BQTIME}(T(n))$ given in terms of a quantum Turing machine by in [FST04]. We argue its equivalence in section 1.3.

Another important class of interest is QMA whose circuit-based definition is as follows.

Definition 1.4 (QMA complexity class). *A promise problem $\mathcal{A}$ is in QMA if and only if there exists a P -uniform quantum circuit family $\{Q_n\}_{n \in \mathbb{N}}$ and a polynomial $p : \mathbb{N} \rightarrow \mathbb{N}$, such that the circuit Q_n takes as input a string $\mathbf{x} \in \mathcal{A}$ and decides on acceptance or rejection of $\mathbf{x}$ as below.*

- If $\mathbf{x} \in \mathcal{A}_{yes} \implies \exists |\psi\rangle \in (\mathbb{C}^2)^{\otimes p(|\mathbf{x}|)}, \Pr[Q \text{ accepts } \{\mathbf{x}, |\psi\rangle\}] \geq 2/3$.
- If $\mathbf{x} \in \mathcal{A}_{no} \implies \forall |\psi\rangle \in (\mathbb{C}^2)^{\otimes p(|\mathbf{x}|)}, \Pr[Q \text{ accepts } \{\mathbf{x}, |\psi\rangle\}] \leq 1/3$.

1.2 QMAH: An infinite hierarchy in QMA

We define individual levels in the QMAH hierarchy as follows.

Definition 1.5 (QMAH_f). *For a specified function $f : \mathbb{N} \rightarrow \mathbb{N}$, a promise problem $\mathcal{A}$ is in QMAH_f , a level of QMAH, if and only if it belongs to QMA as well as $\text{BQSIZE}[f(n)]$.*

Later on, we conditionally prove that if for any two functions f_1 and f_2 , where f_2 asymptotically grows faster than f_1 , then QMAH_{f_1} is strictly contained in QMAH_{f_2} . Alternatively, $f_1 \in o(f_2)$ implies $\text{QMAH}_{f_1} \subset \text{QMAH}_{f_2}$ under the standard assumption of $\text{BQP} \neq \text{QMA}$.

To emphasize the role of choosing two functions, we employ the notion of a parametrized family of functions whose members have an asymptotic growth rate that is faster than any polynomial but slower than any exponential function. The latter restriction on growth with respect to an exponential function is motivated by the fact that QMA is contained in EXP, the set of problems that can be solved by a Turing machine in exponential time. Thus, f_1 and f_2 growing faster than any exponential function would trivially imply $\text{QMAH}_{f_1} = \text{QMAH}_{f_2}$ because any problem in QMA can be decided in $O(2^n)$ time using a classical Turing machine. Similarly, we want the growth of f_1 and f_2 to be asymptotically faster than any polynomial. Otherwise, the problem is trivially in BQP. Thus, the intermediate growth rate of f_1 and f_2 enable us to establish $\text{QMAH}_{f_1} \subset \text{QMAH}_{f_2}$ under a plausible assumption of $\text{BQP} \neq \text{QMA}$.

Definition 1.6 (Indexed family of functions). *We define an indexed family of functions as a set of functions whose elements are functions indexed by integers. Formally, $F := \{f_i \mid \forall i, j \in \mathbb{N}, f_i \in o(f_j) \text{ if } i < j\}$*

We define our hierarchy as follows.

Definition 1.7 (QMAH_i : Level- i of the hierarchy). *For any specified positive integer $i \in \mathbb{N}$, a promise problem $\mathcal{A}$ is in QMAH_i , the i^{th} level of QMAH, if and only if it belongs to QMA as well as $\text{BQSIZE}[f_i]$.*

There are numerous examples of the parametrized family of functions. For the sake of brevity, we illustrate it with the indexed family of quasi-polynomial functions given by $F \equiv \{f_i := 2^{O(\log^i(n))} \mid i \in \mathbb{N}\}$. This naturally instantiate our hierarchy as follows.¹

Definition 1.8 (QMAH_i : Level- i of the hierarchy). *For any specified positive integer $i \in \mathbb{N}$, a promise problem $\mathcal{A}$ is in QMAH_i , the i^{th} level of QMAH, if and only if it belongs to QMA as well as $\text{BQSIZE}[2^{O(\log^i(n))}]$.*

¹it should be noted that for $i = 1$, $f_1 = 2^{O(\log(n))} = \bigcup_{c \in \mathbb{N}} 2^{c \cdot \log(n)} = \bigcup_{c \in \mathbb{N}} 2^{\log(n^c)}$. It can be demonstrated that any polynomial function $g(x) = n^d$ is an element of f_1 . In fact, f_1 is exactly the set of all polynomial functions. Or, every element of f_1 is a polynomial of some degree.

In this setting, the associated QMAH hierarchy is defined to be a union of the classes QMAH_i for all positive integers $i \in \mathbb{N}$.

Definition 1.9 (QMAH Hierarchy). *QMAH is defined as an infinite union of QMAH_i for all positive integer $i \in \mathbb{N}$.*

$$\text{QMAH} \equiv \bigcup_{i \in \mathbb{N}} \text{QMAH}_i.$$

Now we determine the position of the hierarchy with respect to the complexity classes QMA, and BQP.

Theorem 1.1. *The hierarchy $\text{QMAH} \subseteq \text{QMA}$.*

Proof: Definition of each level of the hierarchy dictates $\text{QMAH}_i \subseteq \text{QMA}$ for $\forall i \in \mathbb{N}$. To proof

$$\text{QMAH} \equiv \bigcup_{i \in \mathbb{N}} \text{QMAH}_i \subseteq \text{QMA},$$

we provide the following set-theoretic argument. Take any $x \in \bigcup_{i \in \mathbb{N}} \text{QMAH}_i$. Then, there must be an $i \in \mathbb{N}$ such that $x \in \text{QMAH}_i$. Since $\text{QMAH}_i \subseteq \text{QMA}$ for all $i \in \mathbb{N}$, thus $x \in \text{QMA}$ too. $\square$

Theorem 1.2. *The first level of the hierarchy $\text{QMAH}_1 = \text{BQP}$.*

Proof: The proof of $\text{BQP} \subseteq \text{QMAH}_1$ is due to the fact that BQP is in (i) QMA and (ii) $\bigcup_{c \in \mathbb{N}} \text{BQSIZE}(n^c) = \text{BQSIZE}(2^{O(\log n)})$. For the other direction, $\text{QMAH}_1 \subseteq \text{BQP}$, we use the fact that the set of promise problems in QMA with a polynomial time algorithm is the same as BQP. $\square$

1.3 Defining QMAH through Quantum Turing machine

The definition of the hierarchy in the previous section strictly adheres to the quantum circuit model of quantum computation. In this section, we provide an alternative yet equivalent definition of the hierarchy in the quantum Turing model, which is arguably less prevalent than other models of quantum computing. We start with reviewing the equivalence of these two models proven by [Yao93], and then present the definition of QMAH in terms of the Quantum Turing model of computation.

1.3.1 Known equivalence between Turing and Circuit model of quantum computation

For the sake of completeness, we review the key differences between the quantum Turing and the quantum circuit models, followed by a discussion of the precise technical sense in which the models are equivalent.

Definition 1.10 ([Vaz02]). *A quantum Turing machine (QTM) is defined by a triplet (Σ, S, δ) , where Σ is a finite alphabet with an identified blank symbol $\#$, S is a finite set of states with an identified initial state s_0 and final state s_f , and δ , the quantum finite state control is a function*

$$\delta : S \times \Sigma \mapsto \mathbb{C}^{\Sigma \times S \times D} \quad (1)$$

with $D = \{L, R\}$. The quantum Turing machine has a two-way infinite tape of cells indexed by the integers $\mathbb{Z}$ and a single read/write tape that steps along it.

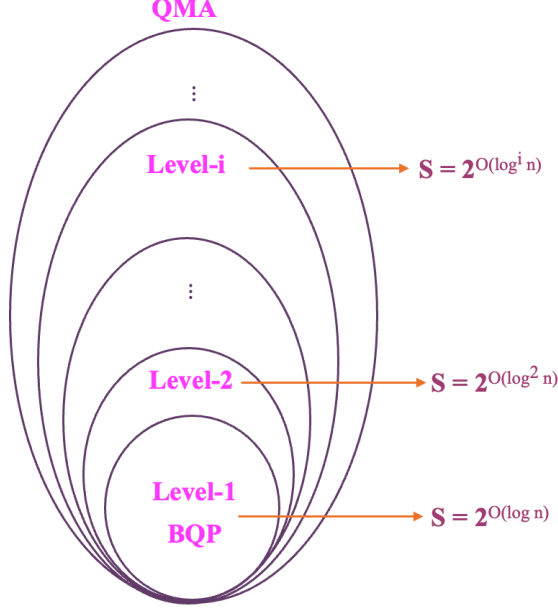


Figure 1: We define an infinite hierarchy in QMA. In one of the instantiations of the hierarchy, the level- i in the hierarchy is a set of promise problems in QMA for which there is a quantum Turing machine that decides the problem using at most $2^{O(\log^i n)}$ steps. As argued in the main text, the choice of the indexed family of functions as $F \equiv \{2^{O(\log^i(n))} | i \in \mathbb{N}\}$ is solely for an illustrative reason. One can choose another indexed family of functions as long as they fulfill two properties characterizing the asymptotic growth of the constituent functions.

A quantum circuit is a generalization of a classical reversible circuit with the following three fundamental changes.

1. Classical registers are replaced by quantum registers.
2. A classical reversible gate set (say, $\{\text{Toffoli}\}$) is replaced with a universal quantum gate set, say $\{\text{Toffoli}, \text{Hadamard}\}$.
3. Quantum measurement in a specified basis (say, standard basis) is deployed at the end of the computation to read out the output.

The equivalence between the quantum Turing model and the quantum circuit model of computation was established by Andrew Yao [Yao93]. We summarize their equivalence by specifying the precise notion of a quantum circuit that simulates a quantum Turing machine.

Let, for any promise problem $\mathcal{A}$, $C_Q(\mathcal{A})$, $D_Q(\mathcal{A})$ be the circuit size and the circuit depth of a quantum circuit Q deciding $\mathcal{A}$, respectively.

Definition 1.11 ((n, t) -simulation of a quantum Turing machine M). *A quantum boolean circuit C_Q with n input register is said to (n, t) -simulate a quantum Turing machine M , if the family of probability distributions p_x for $x \in \{0, 1\}^n$ generated by C_Q is identical to the distributions of the configurations of the machine M after t steps with x as input.*

Lemma 1.1 (Theorem 2 of [Yao93]). *Let M be a quantum Turing machine and n, t be positive integers. Then, there exists a quantum boolean circuit C_Q of size $\text{poly}(n, t)$ that (n, t) -simulates M .*

An inverse of this lemma states any P -uniform quantum circuit can be efficiently simulated by some quantum Turing as follows.

Lemma 1.2 ((n, t) -simulation of a quantum circuit C_Q). *A P -uniform quantum boolean circuit K having n input register can be (n, t) -simulated by a quantum Turing machine M using $O(\text{poly}(|C_Q|))$ tape cells and $O(\text{poly}(|C_Q|))$ number of steps.*

1.4 A Natural Problems at Level- i of QMAH

In this section, we propose a promise problem complete for each level of the QMAH hierarchy, under a polynomial-time deterministic reduction. The problem is a modified variant of the canonical QMA complete problem, the LOCAL HAMILTONIAN (LH) problem [KKR04]. We develop a systematic approach to parametrize the LH problem so that varying parameter values yield a complete problem at various levels of the QMAH hierarchy. It is essential to clarify that the hierarchy as a whole has no complete problems, but individual levels within it do.

1.4.1 Prelude: Guided Local Hamiltonian problem

The design of our complete problems for each level of the hierarchy is inspired by the Guided Local Hamiltonian (GLH(k, γ, δ)) problem [GLG22, CFW22]. It is a variant of LH(γ, δ) where a good guiding state is provided as part of the problem. Guiding state has a dramatic impact on the difficulty of LH(γ, δ), a QMA-complete problem, to a BQP-complete problem. We give a brief exposition of the GLH problem.

The definition of GLH consists of three important terminologies as follows.

Definition 1.12 (Subset State [GKS15]). *For any subset $S \subseteq [2^n]_1$, we define the subset state $|S\rangle \in \mathbb{R}^{2^n}$, as the uniform superposition over the elements of S . More precisely,*

$$|S\rangle := \frac{1}{\sqrt{|S|}} \sum_{i \in S} |i\rangle. \quad (2)$$

Every subset state has a property called support size defined as follows.

Definition 1.13 (Support Size of a subset state). *For a given guided state $|S\rangle$, $\gamma_S := |S|$ is called its Support Size.*

A special subset state with Support Size $\gamma_S \leq \text{poly}(n)$ is called Guided State. It is often denoted as $|\mathbf{u}\rangle$.

Definition 1.14 (Guidance Value of a subset state). *For a given k -LH problem with the Hamiltonian H , the scalar $\gamma(n) \in [0, 1]$ is called the guidance value of a subset state $|S\rangle$*

$$\gamma(n) := \|\Pi_{\text{gs}} |S\rangle\|^2$$

where Π_{gs} is the projection on the subspace spanned by the ground state of H .

This quantifies the quality of a subset state. Having these three terminologies at hand, the problem is defined as follows.

Problem 1.1 ([GLG22] GLH(k, γ, δ) problem). *The problem is a tuple $\langle H; |\mathbf{u}\rangle; (a, b) \rangle$ with the following characteristics. Note: $|\mathbf{u}\rangle$ is a subset state with $|S| \leq \text{poly}(n)$.*

Input: (i) a k -local Hamiltonian matrix $H = \sum_{j=1}^m H_j$ for an n -qubit system such that $\|H\| \leq 1$, (ii) an efficient description of a semi-classical state $\mathbf{u} \in \mathbb{C}^{2^n}$ and (iii) promise gap parameter $a, b \in \mathbb{R}$ such that $|b - a| \geq \delta > 0$.

Promise:

- Fidelity with the ground space $\|\Pi_{\text{gs}} |\mathbf{u}\rangle\| \geq \gamma$.
- Either case (i) $\lambda_0 \leq a$ or (ii) $\lambda_0 \geq b$ holds.

Output:

1. If $\lambda_0(H) \leq a$, output YES.
2. If $\lambda_0(H) \geq b$, output NO.

1.4.2 The parametrized Guided Local Hamiltonian problem (GLH_i)

Parametrized Guided Local Hamiltonian problem is a generalization of the promise problem 1.1. The key difference is the promise attached to the size of the subset state $|S\rangle$. It is no longer bounded by $|S| \leq \text{poly}(n)$.

Problem 1.2 (GLH_p problem). *The problem is a tuple $\langle H; |S_p\rangle; (a, b) \rangle$ with the following characteristics.*

Input: (i) a k -local Hamiltonian matrix $H = \sum_{j=1}^m H_j$ for an n -qubit system such that $\|H\| \leq 1$, (ii) a subset state $|S_p\rangle \in \mathbb{C}^{2^n}$ of size $|S_p| \leq 2^{O(\log^p n)}$ and (iii) promise gap parameter $a, b \in \mathbb{R}$ such that $|b - a| \geq \delta > 0$.

Promise:

- Fidelity with the ground space $\|\Pi_{\text{gs}} |S_p\rangle\| \geq \text{poly}(n)$.
- Either case (i) $\lambda_0 \leq a$ or (ii) $\lambda_0 \geq b$ holds.

Output:

1. If $\lambda_0(H) \leq a$, output YES.
2. If $\lambda_0(H) \geq b$, output NO.

In the next section, we prove that GLH_i is complete for the QMAH_i class under deterministic polynomial-time reduction. For convenience, we will switch to the quantum circuit model of computation, a model equivalent to a quantum Turing machine.

2 Organization of the paper

The remainder of this paper is organized as follows. Section 3 establishes that the problem GLH_i is complete for QMAH_i under deterministic polynomial-time many-one reductions. Section 4 turns from complexity to spectra and asks what, concretely, makes one level harder than another. We begin with a reading of Regev's factoring algorithm [Reg25], in which the quantum Fourier transform serves as a reduction from a problem on the primal lattice to a classically tractable one on the dual lattice, thereby motivating the treatment of spectral concentration as the computational resource. Section 5 converts this analysis into

an algorithm. We give a quantum algorithm for GLH_i whose depth and size scale polynomially in spectral concentration, built from a classical pre-processing step that generates a guiding state from the Pauli normal form of the input Hamiltonian, a quantum phase estimation subroutine, and a classical minimum-finding post-processing step; Its correctness and resource requirements are established in Section 5.2. As anticipated, the running time degrades gracefully as the instance's spectral concentration decreases. Section 6 addresses robustness. The levels are shown to be conditionally robust. Since a strict separation would imply $\text{BQP} \neq \text{QMA}$ and is therefore out of reach unconditionally, we prove that the levels are separated relative to an oracle, adapting the technique of [KF80] with a hybrid oracle built from Simon's problem.

3 Complete Problems with their Proof of Completeness

In this section, we prove that GLH_i is complete for the QMAH_i class under deterministic polynomial-time reduction. The proof of hardness is based on a certain modified version of the Kitaev circuit-to-Hamiltonian construction.

3.1 GLH_i is complete for QMAH_i

As required, the proof of QMAH_i completeness of GLH_i consists of (i) proof of containment of GLH_i in QMAH_i , and (ii) hardness of GLH_i for QMAH_i .

Theorem 3.1. *GLH_i is QMAH_i complete under deterministic polynomial time many-to-one reduction.*

Proof: The proof consists of the following two lemmas that assert the containment of GLH_i in QMAH_i and the hardness of the problem for QMAH_i .

Lemma 3.1 (Containment). *GLH_p is in QMAH_p for all $p \in \mathbb{N}$.*

Proof: This follows from the definition of the problem GLH_p . The problem has a guiding state $|S_p\rangle \leq 2^{O(\log^p n)}$. Thus, the quantum phase estimation algorithm can be used to decide the membership in QMAH_i . $\square$

Lemma 3.2 (Hardness). *For any fixed $p \in \mathbb{N}$, a promise problem $A \in \text{QMAH}_p$ reduces deterministically in polynomial-time to an instance of GLH_p problem, i.e.*

$$\text{QMAH}_p = \{A | A \leq_p^m \text{GLH}_p\}.$$

If a problem is both (i) in QMAH_i and (ii) hard for QMAH_i , then it is a complete problem for the class. $\square$

3.1.1 Proof of containment of GLH_i in QMAH_i

The proof of Lemma 3.1 goes as follows. Notice GLH_i satisfies both requirements for being a member of QMAH_i . It is in (i) QMA and (ii) there exists a quantum algorithm that demonstrates $\text{GLH}_i \in \text{BQTIME}(2^{O(\log^i n)})$. The witness is the subset state $|S\rangle \in \mathbb{C}^{2^n}$ that has a good overlap with the ground state of the given Hamiltonian. We have provided a quantum algorithm with gate complexity (circuit size) of $2^{O(\log^i n)}$ in section 5.1.

3.1.2 Proof of hardness of GLH_i for QMAH_i

Our proof of Lemma 3.2 goes as follows. There are six steps of the reduction that take a QMAH_i circuit and output an instance of GLH_i . After that, we analyze the completeness and soundness of the reduction procedure.

Step I: Given an instance of QMAH_i problem, say $\langle U, a, b \rangle$, reduce it to an instance of UQMAH_i , say $\langle \hat{U}, \hat{a}, \hat{b} \rangle$. Let the unique witness for the former case be $y^* \in \{0, 1\}^w$ where $w = O(\log^p n)$. Note: parameters a and b are the promise gap for the problem.

Step II: Use the CNOT-trick to modify the β_p - QMAH_i verification circuit to force the witness to be classical.

Step III: Use Marriott and Watrous Strong error reduction on the output (verification) circuit of Step II. Let the output of this step be $\langle \tilde{U}, \tilde{c}, \tilde{s} \rangle$, where $\tilde{c} = 1 - 2^r$ and $\tilde{s} = 2^r$ for some $r \in \text{poly}(n)$, and circuit depth be $\tilde{T} = \text{poly}(n)$. Assume, the qubit layout for verification circuit $\tilde{U}$ is (i) input x in register A , (ii) the witness y^* in register W , and (iii) $\text{poly}(n)$ sized register B for ancilla.

Step IV: Use modified Feynman-Kitaev (FK) Circuit-to-Hamiltonian construction, given by [DGF22], to convert $\langle \tilde{U}, \tilde{c}, \tilde{s} \rangle$ into local Hamiltonian. The motivation for using the modified FK is the presence of a tunable parameter, ϵ , which shifts the energy level according to the acceptance probability of the verification circuit. This is crucial to get the final form of the Hamiltonian of Step V that has the desired β_p Guidable State. To develop an intuition, let x be the input and $y^* \in \{0, 1\}^{\log^p(n)}$ be the witness associated with $\langle \tilde{U}, \tilde{c}, \tilde{s} \rangle$. The output Hamiltonian from the DFG construction will take the following form.

$$H_{\text{FK}}^x = H_{\text{in}} + H_{\text{clock}} + H_{\text{prop}} + \epsilon \cdot H_{\text{out}} \quad (3)$$

The ground state of $H_0 = H_{\text{in}} + H_{\text{clock}} + H_{\text{prop}}$ is called **History-State**

$$|\eta(y^*)\rangle = \frac{1}{\sqrt{\tilde{T} + 1}} \sum_{t=0}^{\tilde{T}} \tilde{U}_t \dots \tilde{U}_1 |y^*\rangle |0\rangle |\tilde{t}\rangle. \quad (4)$$

If the verification circuit accepts $(x, |y^*\rangle)$ with p probability, then the **History State** has the following energy with respect to the reduced Hamiltonian H_{FK}^x .

$$\langle \eta(y^*) | H_{\text{FK}}^x | \eta(y^*) \rangle = \epsilon \cdot \frac{1 - p}{\tilde{T} + 1} \quad (5)$$

The left-hand side includes the reduced Hamiltonian, while the right-hand side contains the verification circuit parameters. The parameter ϵ can be used as a knob to adjust the relationship between them. In its full generality, the DFG construction can be used to gain fine control over the relation between the acceptance probabilities of the circuit and the low-energy sector of the Hamiltonian, as formalized in the next lemma.

Lemma 3.3. (Small penalty clock construction) [DGF22]: Let U_n be a quantum verification circuit for inputs x , $|x| = n$, where U_n consists of $T = \text{poly}(n)$ gates from some universal gate-set using at most 2-local gates. Denote $P(|\psi\rangle)$ for the probability that U_n

accepts $(\mathbf{x}, |\psi\rangle)$, and let H_{FK}^x be the corresponding 3-local Hamiltonian from the circuit-to-Hamiltonian mapping in [KKR04] with a ϵ -factor in front of H_{out} , as in

$$H_{FK}^x = H_{\text{in}} + H_{\text{clock}} + H_{\text{prop}} + \epsilon \cdot H_{\text{out}}. \quad (6)$$

then for all $\epsilon \leq c \cdot T^3$ for some $c > 0$, we have that low-energy subspace S_ϵ of H , that is

$$S_\epsilon = \text{span}|\Phi\rangle : \langle \Phi | H | \Phi \rangle \leq \epsilon \quad (7)$$

has that its eigenvalues λ_i satisfy

$$\lambda_i \in \left[\epsilon \frac{1 - P(|\psi\rangle)}{T + 1} - O(T^3 \epsilon^2), \epsilon \frac{1 - P(|\psi\rangle)}{T + 1} + O(T^3 \epsilon^2) \right], \quad (8)$$

where $\{|\psi_i\rangle\}$ are the eigen-states of the verification circuit corresponding to $\tilde{U}$.

The above lemma will be used to fix the parameters arising in the next step, such that the completeness and soundness property of the reduction is preserved.

Step V: To have the desired β_p Guided State, the Hamiltonian of Step IV needs to be block encoded using one additional single qubit.

$$H^x = H_{\text{yes}} \otimes |0\rangle\langle 0|_D + H_{\text{no}} \otimes |1\rangle\langle 1|_D \quad (9)$$

where $H_{\text{yes}} = H_{FK}^x$ and

$$H_{\text{no}} = \sum_{i=0}^{R-1} |1\rangle\langle 1|_i + bI \quad (10)$$

where R is the total size of the registers used for $H_{\text{yes}} = H_{FK}^x$. The two desired advantages of this block encoding are-

- (i) H_{no} has a unique ground state $|0\dots 0\rangle$ with the energy b . Thus, in NO-case, $|u_{\text{no}}\rangle = |0\dots 0\rangle_{FK} |1\rangle_D$ is the ground state with the energy b .
- (ii) The guiding state for YES-case will be the basis state

$$|u_{\text{yes}}\rangle = |x\rangle_A |y^*\rangle_W |0\dots 0\rangle_B |0\rangle_C |0\rangle_D \quad (11)$$

which satisfies $(\langle \eta(y^*) | \langle 0 |_D) | u_{\text{yes}} \rangle = 1/\sqrt{\tilde{T} + 1} = O(1/\text{poly}(n))$. Notice, $|u_{\text{yes}}\rangle$ is a β_p -Guiding State as the state can be partitioned into two parts as

$$|u_{\text{yes}}\rangle = \left(|y^*\rangle_W \right)_I \left(|x\rangle_A |0\dots 0\rangle_B |0\rangle_C |0\rangle_D \right)_{II} \quad (12)$$

where part I is of $O(\log p n)$ size while, part II is of $\text{poly}(n)$ size. Notice, part II can be computed efficiently by Arthur, while part I indispensably requires Merlin's assistance if $\text{BQP} \neq \text{QMAH}_i$.

Step VI: This step is to increase the fidelity parameter $\gamma(n)$. We can increase the fidelity of $|u_{\text{yes}}\rangle$ with the ground state from $1/\text{poly}(n)$ to $1 - 1/\text{poly}(n)$ using the pre-idling trick, that is, appending a $\text{poly}(n)$ identities to the actual unitary circuit $\tilde{U}$ to get $\tilde{U}_{\text{new}} = I \dots I \tilde{U}$. thus the total depth of the circuit increases from $\tilde{T}$ to $\tilde{T}_{\text{new}} = \tilde{T} + N$, where $N \in \text{poly}(n)$. The new guiding state is given by

$$|u_{\text{yes}}^{\text{new}}\rangle = \left(\frac{1}{\sqrt{\tilde{T} + N}} \sum_{t=0}^N |x\rangle_A |y^*\rangle_W |0\dots 0\rangle_B |t\rangle_C |0\rangle_D \right) \quad (13)$$

or, in the β_p Guiding state form

$$|u_{\text{yes}}^{\text{new}}\rangle = \left(\frac{1}{\sqrt{\tilde{T} + N}} \sum_{t=0}^N |x\rangle_A |0\dots 0\rangle_B |t\rangle_C |0\rangle_D \right) \otimes (|y^*\rangle_W). \quad (14)$$

This has the following fidelity with History state

$$|\langle \eta(y^*) | \langle 0 |_D \rangle | u_{\text{yes}}^{\text{new}} \rangle|^2 = \frac{N}{N + \tilde{T} + 1}. \quad (15)$$

Now we use the proven fact that the fidelity of History state with the actual ground state ($|gs\rangle$) is exponentially close to one, that is, $\geq 1 - 2^r$ for some $r \in \text{poly}(n)$. Thus, the lower bound on the fidelity of Guiding state with the actual ground state is-

$$|\langle u_{\text{yes}}^{\text{new}} | gs \rangle|^2 \geq 1 - \left(\sqrt{1 - |\langle \eta(y^*) | \langle 0 |_D \rangle | u_{\text{yes}}^{\text{new}} \rangle|^2} + \sqrt{1 - |\langle gs | u_{\text{yes}}^{\text{new}} \rangle|^2} \right) \geq 1 - \frac{1}{r(n)} \quad (16)$$

where, the polynomial $r \in \text{poly}(n)$ get fixed once we suitably choose the polynomial $N \in \text{poly}(\tilde{T})$ of eqn.15 so that it obeys the eqn. 20.

Completeness argument: Using the lemma 3.3 mentioned in StepIV, the optimal choice for parameter b (eqn.10) and ϵ (eqn.6) turns out to be $b := O(1/\tilde{T}^7)$ and $\epsilon := O(1/\tilde{T}^5)$. We give a brief argument for this choice. Notice, lemma 3.3 asserts the eigenvalue corresponding to unique witness y^* is upper bounded as

$$\lambda(y^*) \leq \epsilon \frac{2^{-r}}{\tilde{T} + 1} + O(\tilde{T}^3 \epsilon^2). \quad (17)$$

While, for any other $y \neq y^*$, the corresponding eigenvalue would be lower bounded as

$$\lambda(y) \geq \epsilon \frac{1 - 2^{-r}}{\tilde{T} + 1} - O(\tilde{T}^3 \epsilon^2) = \Omega\left(\frac{1}{\tilde{T}^6}\right). \quad (18)$$

Now, it can be verified that the spectral gap for the reduced Hamiltonian would be lower bounded as follows (if we take $\epsilon := O(1/\tilde{T}^5)$).

$$\Delta(H_{\text{yes}}) \geq \epsilon \cdot \frac{1 - 2^{-r+1}}{\tilde{T} + 1} - O(\tilde{T}^3 \epsilon^2) = \Omega\left(\frac{1}{\tilde{T}^6}\right) \quad (19)$$

This is used to derive the lower bound on the fidelity of $|u_{\text{yes}}\rangle$ with the ground state, say $|gs\rangle$, of the reduced Hamiltonian as

$$|\langle u_{\text{yes}} | gs \rangle|^2 \geq 1 - \left(\sqrt{1 - |\langle \eta(y^*) | \langle 0 |_D \rangle | u_{\text{yes}} \rangle|^2} + \sqrt{1 - |\langle gs | u_{\text{yes}} \rangle|^2} \right) \geq 1 - \Omega\left(\frac{1}{\tilde{T}}\right) \quad (20)$$

Soundness argument: In NO-case, the choice of the parameter $b := O(1/\tilde{T}^7)$ implies the energy of any possible Guiding state is at least $1/\text{poly}(n)$. This completes the proof of QMAH _{i} hardness of GLH _{i} for all $i \in \mathbb{N}$. $\square$

4 Spectral interpretation of GLH_i

The notion of Fourier spectral concentration is well-defined for Boolean functions and has been utilized for learning certain Boolean functions [O'D14, PPR22]. Here, we extend this analysis to the local Hamiltonian problem, which can be viewed as a quantum generalization of a Boolean function [MO08]. In this section, we define it and discuss its utility for studying the complexity of the ground state in the local Hamiltonian problem.

Before giving the definitions, we take a detour to Regev's recent algorithm for the factorization problem, which implicitly relies on the Fourier spectrum being sufficiently concentrated to yield a polynomial-time quantum algorithm.

4.1 Background of Regev's Factorization Algorithms

The problem of integer factorization admits an efficient quantum algorithm [Sho99, Reg25]. In this section, we reinterpret Regev's algorithm from an alternative perspective. The key motivation is to find a unifying characteristic property in any problem that has an efficient quantum algorithm, i.e., any BQP problem. Let $N \leq 2^n$ be an n -bit number. Let $d = \lfloor \sqrt{n} \rfloor$ and $b_1, \dots, b_d$ be some small $O(\log n)$ -bit integers (that is, b_i is the i -th prime number) and let $a_i = b_i^2 \bmod N$.

Let D be the power of 2 in $[2\sqrt{d} \cdot R, 4\sqrt{d} \cdot R]$, where $R = 2^{C+2+o(1)}\sqrt{n}$. The constant C is prescribed by the number-theoretic conjecture assumed by Regev [Reg25].

Lattice(group): Consider the following d dimensional lattice:

$$\mathcal{L} = \left\{ (z_1, \dots, z_d) \in \mathbb{Z}^d : \prod_{i=1}^d a_i^{z_i} \equiv 1 \bmod N \right\} \text{ and,}$$

$$\mathcal{L}_0 = \left\{ (z_1, \dots, z_d) \in \mathbb{Z}^d : \prod_{i=1}^d b_i^{z_i} \equiv \pm 1 \bmod N \right\} \subseteq \mathcal{L}.$$

And its dual lattice is defined as:

$$\mathcal{L}^* = \{ y \in \mathbb{R}^d : \langle x, y \rangle \in \mathbb{Z} \ \forall x \in \mathcal{L} \} \supseteq \mathbb{Z}^d.$$

Key idea: The central idea behind the [Reg25] algorithm is to find a vector $z = (z_1, \dots, z_d) \in \mathcal{L}/\mathcal{L}_0$. This suffices because, using z , we can construct $b = \prod_{i=1}^d a_i^{z_i} \bmod N$. Since $z \in \mathcal{L}$, b must be a non-trivial square root of 1 modulo N . This implies that N must divide $(b-1)(b+1)$. Now, $\text{GCD}(b-1, N)$ will be a non-trivial divisor of N .

Theorem 4.1 ([Reg25]). *Let N be an n -bit number and assume that for $d = \sqrt{n}$ and $O(\log(n))$ bit numbers $b_1, \dots, b_d$, there exists a vector in $\mathcal{L}/\mathcal{L}_0$ of the norm at most $T = \exp(O(\sqrt{n}))$. Then, there is a classical polynomial-time algorithm that outputs a non-trivial factor of N using $\sqrt{n} + 4$ calls to a quantum circuit of size $O(n^{3/2} \log(n))$.*

4.1.1 Quantum and Classical Subroutines in the Regev's Algorithm

Regev's algorithms can be segmented into two parts: (i) a quantum circuit to compute the Fourier transform of the primal lattice $\mathcal{L}$ to the dual lattice $\mathcal{L}^*$. The circuit is run

polynomially many times, that is, $\sqrt{n}+4$ times, and independently sampled. (ii) A classical post-processing method that acts on the sampled set to recover a basis for vectors in $\mathcal{L}$ with a maximum norm of $T = 2^{C\sqrt{n}}$.

4.1.2 The quantum subroutine

1. Construction of the superposition state over vectors z :

$$\sum_{z \in \{-D/2, \dots, D/2\}^d} \exp(-\pi \|z\|^2 / R^2) |z\rangle$$

2. Coherent evaluation of the function:

$$|z\rangle |0^M\rangle \mapsto |z\rangle |\Pi_{i=1}^d a_i^{z_i + D/2} \bmod N\rangle |\psi\rangle$$

3. Quantum Fourier transform: Measure the register storing $\Pi_{i=1}^d a_i^{z_i + D/2} \bmod N$ to collapse

the $|z\rangle$ register to a superposition over some arbitrary coset from $\mathbb{Z}^d / \mathcal{L}$. This is followed by the uncomputing of state $|\psi\rangle$. After this, the quantum Fourier transform (QFT) over the group $\mathbb{Z}_D^d$ to the $|z\rangle$ register. This transforms the quantum state into the dual lattice group $\mathcal{L}^* / \mathbb{Z}^d$. 4. Sampling: The final measurement of the quantum state after QFT would

uniformly give us a vector close to a point in the dual lattice. To be more precise, with the probability of $1 - 1/\text{poly}(d)$, we obtain a noisy sample of the form $w_i = v_i + \delta_i$. Here v_i is a uniform sample from $\mathcal{L}^* / \mathbb{Z}^d$ and δ_i is an error of magnitude at most $2^{-(C+o(1))n/d}$.

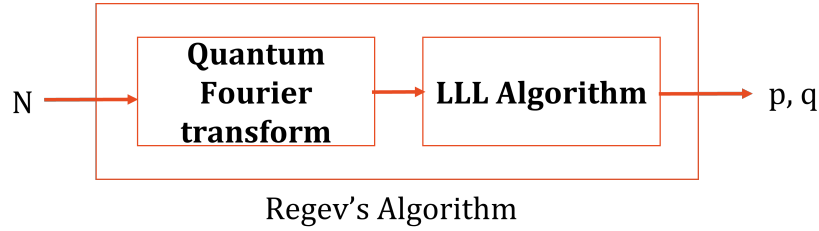


Figure 2: Quantum and Classical parts in Regev's algorithm

4.1.3 The classical post-processing

1. Lattice Basis reduction algorithm: The noisy sample $w_i = v_i + \delta_i$ is used to recover v_i using the classical Lenstra–Lenstra–Lovász lattice basis reduction algorithm [LLL82].
- 2.

Maximum size of the sampled set : The LLL algorithm has the stringent requirement that the sampled set must contain the basis-forming set for the underlying lattice. Otherwise, the algorithm will output some garbage.

4.1.4 Abundance of the Basis Forming Sample Sets

A lemma due to Carl Pomerance [Pom02] asserts that if we uniformly sample $d+4$ elements from a group (with replacement), then it contains the basis (or generator) of the group with at least $1/2$ probability. Formally,

Lemma 4.1 ([Pom02]). *Suppose that G is a finite Abelian group with minimal number of generators r . Then, when choosing elements from G independently and uniformly, the expected number of elements needed to generate G is less than $r + \sigma$, where $\sigma = 2.118456\dots$*

Theorem 4.2. *In the setting of the above lemma 4.1, $r + 4$ uniformly random elements of G generate G with probability at least $1/2$.*

4.1.5 Quantum Fourier transform as quantum reduction algorithm

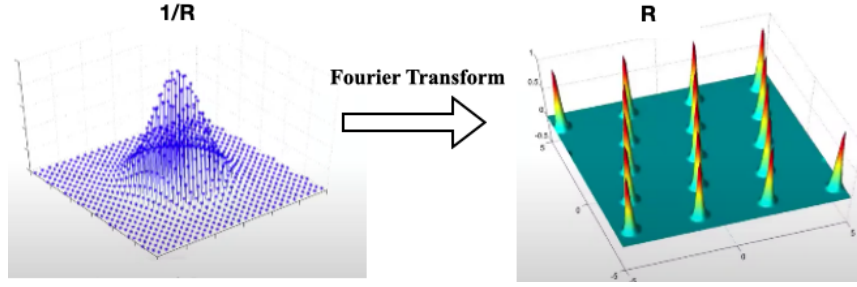


Figure 3: (Fourier transform of the primal lattice to the dual lattice) QFT can be understood as facilitating a quantum reduction of a problem defined on the primal lattice to a problem defined on the dual lattice. The latter problem is classically easy, whereas the former is known to have no efficient classical algorithm. Note: Symbol R represents the width of the associated Gaussian peaks as discussed in Subsection 4.1.1.

We can interpret the role of the quantum Fourier transform as doing a quantum Turing reduction of the problem defined on the primal lattice, say problem I, to the problem defined on the dual lattice, say problem II. Problem I is to compute a non-trivial vector in the primal lattice, while problem II is to compute the basis of the primal lattice using a noisy sample vector from the dual lattice. Problem II is classically tractable, while it is unknown if Problem I is classically tractable or not. Thus, the quantum Fourier transform reduces a classically hard problem to a classically easy problem.

4.2 Spectral Concentration γ_i of GLH_i

We recall that the definition of GLH_i includes a subset state as described below.

Problem 4.1 (GLH_i problem). *The problem is a tuple $\langle H; |S\rangle; (a, b) \rangle$ with the following characteristics.*

Input: (i) a k -local Hamiltonian matrix $H = \sum_{j=1}^m H_j$ for an n -qubit system such that $\|H\| \leq 1$, (ii) a subset state $|S\rangle \in \mathbb{C}^{2^n}$ of size $|S| \leq 2^{O(\log^i n)}$ and (iii) promise gap parameter $a, b \in \mathbb{R}$ such that $|b - a| \geq \delta > 0$.

Promise:

- Fidelity with the ground space $\|\Pi_{\text{gs}} |S\rangle\| \geq \text{poly}(n)$.
- Either case (i) $\lambda_0 \leq a$ or (ii) $\lambda_0 \geq b$ holds.

Output: Output YES if $\lambda_0(H) \leq a$, and output NO if $\lambda_0(H) \geq b$.

We defined the spectral concentration of an input tuple as follows.

Definition 4.1 (Spectral concentration of a GLH_i instance). *Give an instance $\langle H; |S\rangle; (a, b) \rangle \in \text{GLH}_i$. Its spectral concentration is defined as the inverse of the size of the set S . Or,*

$$\gamma_S := \frac{1}{|S|}.$$

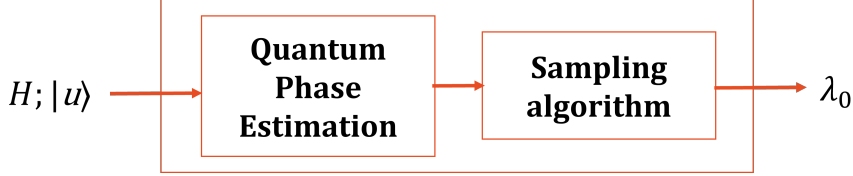


Figure 4: Quantum and Classical components in the algorithm

4.2.1 The quantum algorithm

The algorithm finds the minimum eigenvalue of the given local Hamiltonian and the guided state. First, we use the quantum phase/eigenvalue estimation algorithm (QPE) [Kit95]. QPE is run a polynomial number of times and is sampled independently. Secondly, classic post-processing on the sampled set yields the minimum eigenvalue λ_0 . The formal details of the QPE algorithm are as follows:

Remark: The problem of estimating the phase of the unitary operator can be used to estimate an eigenvalue of the Hermitian operator due to the following lemma.

Theorem 4.3. *If for a Hermitian matrix $H \in \mathbb{C}^{N \times N}$ and $\mathbf{u} \in \mathbb{C}^N$,*

$$H\mathbf{u}_i = \lambda_i \mathbf{u}_i ; \quad \forall i \in [N]_0$$

then the following eigenvalue relation holds for the unitary matrix $U = e^{iH}$

$$U\mathbf{u}_i = e^{i\lambda_i} \mathbf{u}_i ; \forall i \in [N]_0$$

4.2.2 Quantum subroutine: quantum phase estimation QPE

Context for QPE: We have a sparse unitary operator on which a controlled- U^j operation can be efficiently performed for $j \in \mathbb{Z}$. Furthermore, we are given an eigenstate $|\mathbf{u}\rangle$ of U with eigenvalue $e^{2\pi i \phi_u}$. The task is to find $\tilde{\phi}_u$ such that $\|\tilde{\phi}_u - \phi_u\| \leq \epsilon$.

4.2.3 Algorithmic Procedure for QPE

1. Prepare the initial state:

$$|0\rangle^m |\mathbf{u}\rangle |\mathbf{x}\rangle$$

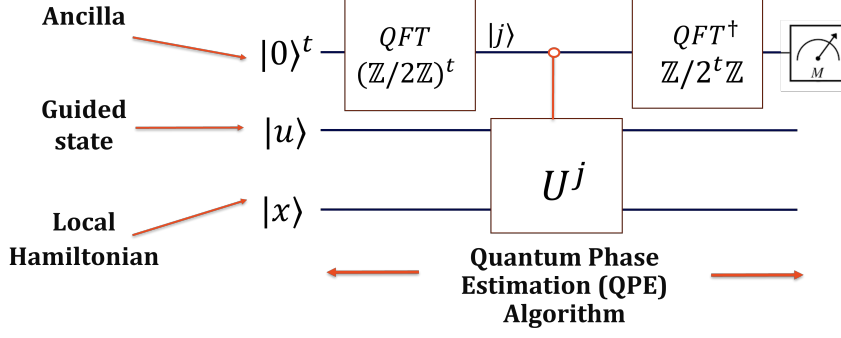


Figure 5: (Quantum Phase/Eigenvalue Estimation) A guided state $|\mathbf{u}\rangle$ and the description of the local Hamiltonian $\mathbf{x}$ is provided as input. The unitary $U = e^{i\mathbf{x}}$. Two different types of quantum Fourier transform have been employed: (i) a quantum Fourier transform over the Boolean group $(\mathbb{Z}/2\mathbb{Z})^2$, also called the Hadamard transform, and (ii) a quantum Fourier transform over the group $\mathbb{Z}/2^n\mathbb{Z}$.

2. Create the superposition:

$$|0\rangle^m |\mathbf{u}\rangle |\mathbf{x}\rangle \rightarrow \frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} |j\rangle |\mathbf{u}\rangle |\mathbf{x}\rangle$$

3. Apply controlled- U^j :

$$\frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} |j\rangle |\mathbf{u}\rangle |\mathbf{x}\rangle \rightarrow \frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} |j\rangle U^j |\mathbf{u}\rangle |\mathbf{x}\rangle = \frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} e^{2\pi i j \phi_u} |j\rangle |\mathbf{u}\rangle |\mathbf{x}\rangle$$

4. Apply the inverse Fourier transform:

$$\frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} e^{2\pi i j \phi_u} |j\rangle |\mathbf{u}\rangle |\mathbf{x}\rangle \rightarrow |\phi_u\rangle |\mathbf{u}\rangle |\mathbf{x}\rangle$$

5. Measure the first register and output the number:

$$|\phi_u\rangle |\mathbf{u}\rangle |\mathbf{x}\rangle \rightarrow \tilde{\phi}_u$$

4.2.4 Resource and running time analysis for QPE

Lemma 4.2. *Quantum eigenvalue estimation lemma: Let $H = \sum_{i=1}^m H_i$ be a 2-local Hamiltonian acting on n qubits, with eigenvectors $|\psi_j\rangle$ and the corresponding eigenvalues $\lambda_j \in [0, 1]$. Then there is a quantum circuit that, given an input eigenvector $|\psi_j\rangle$, will output (with probability $\geq p$) an ϵ -approximation of λ_j (that is, $\tilde{\lambda}_j$ such that $|\tilde{\lambda}_j - \lambda_j| \leq \epsilon$). The depth of the quantum circuit is*

$$d \leq \text{poly}\left(m, \frac{1}{\epsilon}, \frac{1}{p}\right). \quad (21)$$

4.2.5 Classical subroutine: the sampling algorithm

Each measurement of QPE gives an eigenvalue of the input Hamiltonian. Since our goal is to find the minimum eigenvalue, we need to collect a sufficiently large sample to ensure it contains the minimum eigenvalue with high probability. Thus, we need to run the QPE circuit multiple times. The upper bound on the repetition of QPE is estimated using the following theorem.

Theorem 4.4. *The maximum number of times the QPE algorithm needs to be repeated to get a sample set from which λ_0 (min. eigenvalue) can be extracted with the probability of $1/2 + \delta_p$ is*

$$O(\log(1/\delta_p)).$$

Proof: Since the probability of getting λ_0 in the first trial is greater than γ . Then,

$$\Pr[\text{Not getting } \lambda_0 \text{ in the first trial}] \leq 1 - \frac{1}{\gamma}$$

Similarly,

$$\Pr[\text{Not getting } \lambda_0 \text{ in the } t\text{-th trial}] \leq 1 - \left(\frac{1}{\gamma}\right)^t$$

As per our requirement, we want

$$1 - \left(\frac{1}{\gamma}\right)^t \geq \frac{1}{2} + \delta_p \implies \frac{1}{2} - \delta_p \geq \left(\frac{1}{\gamma}\right)^t. \quad (22)$$

$$t \leq \frac{\log\left(\frac{1}{2} - \delta_p\right)}{\log \frac{1}{\gamma}}. \quad (23)$$

4.2.6 The Spectral Density Analysis

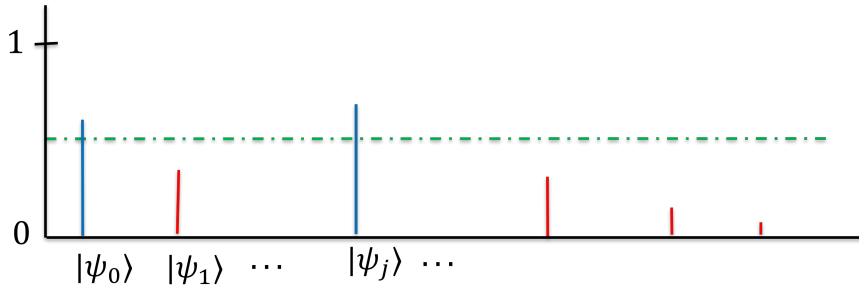


Figure 6: (Spectral interpretation) The Height of a peak represents the fidelity of the guided state with an eigenstate of the Hamiltonian H . The blue peaks crossing above the horizontal green line correspond to the eigenstates that have substantial overlap with the guided state. The eigenvalue associated with these 'blue' eigenstates will appear more frequently in the sample collected via the QPE algorithm.

Due to the spectral theorem, any Hermitian matrix $H \in \mathbb{C}^{N \times N}$ can be decomposed into its eigenbasis as:

$$H = \sum_{i=0}^{N-1} \lambda_i |\psi_i\rangle \langle \psi_i|$$

For any normalized vector $|\mathbf{u}\rangle \in \mathbb{C}^N$, the spectral measure induced by the matrix H and the vector $|\mathbf{u}\rangle$ is a probability distribution on the spectrum of H such that the probability of getting eigenvalue λ_i occurs with

$$\Pr[\text{measuring } \lambda_i] \geq |\langle \psi_i | \mathbf{u} \rangle|^2. \quad (24)$$

Typically, we don't have direct access to the ground state of a Hamiltonian; instead, we have access to a state that overlaps with it. However, this can be used to compute the minimum eigenvalue with a success probability depending on the extent of the overlap.

Lemma 4.3. *Suppose we have been given a state*

$$|\mathbf{u}\rangle = \sum_{i=0}^{N-1} \alpha_i |\psi_i\rangle \quad (25)$$

instead of the ground state $|\psi_0\rangle$ for computing using the QPE algorithm. Then the probability of measuring $\lambda_0(H)$ when the input to the QPE is $(H, |\mathbf{u}\rangle)$ is lower bounded by

$$\Pr[\text{measuring } \lambda_i] \geq |\alpha_0|^2(1 - \epsilon) \quad (26)$$

where $(1 - \epsilon)$ is the probability of success of the phase estimation.

4.2.7 Abundance of low-energy states and the success probability of QPE

QFT plays the role of a change of basis from a standard basis to the eigenbasis of the Hamiltonian matrix. The standard basis consists of the set $\mathcal{B} = \{|\mathbf{z}\rangle \mid \mathbf{z} \in \{0, 1\}^n\}$ while the set of eigenbasis of the Hamiltonian H is the set $\mathcal{E} = \{|\psi\rangle \mid H|\psi\rangle = \lambda|\psi\rangle; |\psi\rangle \in \mathbb{R}^{2^n}\}$.

Stroeks et al. have shown that if the guiding state has good overlap with the $O(1)$ eigenstates, then there is a randomized algorithm to solve the guided local Hamiltonian problem. We propose a new variant of the guided local Hamiltonian problem that quantifies the number of eigenstates with good overlap with the guiding vector as a computational resource.

4.3 An Algorithm to Estimate γ_i

4.3.1 Relating Guidance Value and Quantum Circuit Depth

It is anticipated that a problem with a higher guidance value will be solved using a relatively shallow quantum circuit compared to a problem with a lower guidance value. Now, we will precisely quantify the dependency of guidance value (γ) on circuit depth ($d(n)$) through the following analysis.

Theorem 4.5. *There exists a method to generate a guided state $|\mathbf{u}\rangle$ such that the value of the guidance parameter scales as*

$$\gamma := |\langle \mathbf{u} | \psi_{\text{hist}} \rangle|^2 \geq \frac{N}{d(n) + N + 1}. \quad (27)$$

where, $N = O(d(n))$.

Proof: It follows from the hardness result for the GLH_i problem for the class β_p -QCMA. The key observation is that, in the worst-case scenario, the maximally mixed state is the optimal choice as the guiding state. It has the following practical corollaries of interest.

Corollary 4.1. *If $d(n) = \text{poly}(n)$, then there exists a guiding state $|\mathbf{u}\rangle$ with*

$$\gamma := |\langle \mathbf{u} | \psi_{\text{hist}} \rangle|^2 \geq \frac{1}{\text{poly}(n)}.$$

Corollary 4.2. *If $d(n) = 2^{c \cdot \log^p(n)}$, then there exists a guiding state $|\mathbf{u}\rangle$ with*

$$\gamma := |\langle \mathbf{u} | \psi_{\text{hist}} \rangle|^2 \geq \frac{1}{2^{\text{polylog}(n)}}.$$

Corollary 4.3. *If $d(n) = 2^{\text{poly}(n)}$, then there exists a guiding state $|\mathbf{u}\rangle$ with*

$$\gamma := |\langle \mathbf{u} | \psi_{\text{hist}} \rangle|^2 \geq \frac{1}{2^{\text{poly}(n)}}.$$

4.4 Relating γ_i to level i of QMAH

We present two criteria under which some variant of LH problems are either classically or quantumly easy. This will illustrate why the presence of certain types of Pauli (string) Hamiltonians² with degree $d \leq 2$ with a frequency less than a certain threshold is tractable. This will help build intuition and the proof of correctness.

Classically tractable case: For the classically tractable case, we start with the one local Hamiltonian, which is known to be in P.

Problem 4.2 ($\{X, Y, Z\}$ -Hamiltonians). *Input: A one local Hamiltonian on n -qubit system which takes the following form*

$$H = \sum_{k \in [n]} \alpha_k X_k + \beta_k Y_k + \gamma_k Z_k \quad (28)$$

Promise : *Either, (i) $\lambda_0(H) \leq a$, or (ii) $\lambda_0(H) \geq b$. Along with the promise gap of $b - a \geq 1/\text{poly}(n)$.*

A polynomial-time algorithm exists for the problem. We mention it for completeness.

Theorem 4.6. *1-Local Hamiltonian is in P.*

Proof: There are two methods to construct this. First, construct the ground state, which is guaranteed to be a product state over n qubits.

$$|\psi\rangle = \bigotimes_{i=1}^n |\psi_i\rangle \quad (29)$$

Now, each one-particle state $|\psi_i\rangle$ can be identified, which minimizes the associated 1-local Hamiltonian terms (in which the i th particle participates). Each qubit can be present in a maximum of three clauses/terms. Thus, we need to run the checking process at most a polynomial number of times.³

The second proof evokes a result from matrix theory. For 1-local Hermitian matrices, there always exists a 1-qubit unitary matrix $U_{2 \times 2}$ such that they $U^{\otimes n}$ can simultaneously

²The matrix $Z_1 \otimes Z_2 \otimes I_3 \otimes I_4 \otimes Z_5$ is a five-qubit Pauli (string) Hamiltonian with the Pauli degree three.

³**Note:** the same technique won't help give a polynomial time algorithm for the 2-Local Hamiltonian problem. The reason is that each qubit might share a clause/term with $(n-1)$ other qubits. The same method would require an exponential number of checks, in the worst case.

diagonalize the input 1-Local Hamiltonian H . Once diagonalized, one can easily find the minimal eigenvalue by inspecting the minimum of the diagonal entries. $\square$

Next, we introduce an **NP-complete** version of 2-local Hamiltonian problem due to [Bar82]. The reason for the Hardness is due to the appearance of **Complexity generator** terms in high frequency.

Problem 4.3 ($\{ZZ, Z\}$ -Hamiltonians). *Input: A two local Hamiltonian on n -qubit system which takes the following form*

$$H = \sum_{i,j \in [m]} \alpha_{i,j} Z_i \otimes Z_j + \sum_{k \in [n]} \beta_k Z_k \quad (30)$$

Promise : *Either, (i) $\lambda_0(H) \leq a$, or (ii) $\lambda_0(H) \geq b$. Along with the promise gap of $b - a \geq 1/\text{poly}(n)$.*

The problem is **NP-complete** if the parameter $m = \text{poly}(n)$. We prove that the problem is in **P** if we restrict the parameter.

Theorem 4.7. *The problem 4.3 is in **P** if parameter $m = O(\log(n))$.*

Proof : The complexity of the problem is due to the appearance of the $Z_i \otimes Z_j$ terms in the Hamiltonian. We (informally) call it classical **complexity generating** terms. The 1-local term does not add to the complexity. Restricting the parameter $m = O(\log(n))$ implies the number of variables appearing in **complexity generating** terms is not too large. Thus, the search space is upper bounded by $\text{poly}(n)$. The reason is that the number of possible basis states/assignments over $m = O(\log(n))$ many qubits/variables is $2^m = \text{poly}(n)$. Therefore, one can check each $m = O(\log(n))$ possibility, where each check will require at most $\text{poly}(n)$ time.⁴ It should be noted that the one-local Hamiltonian can be managed using the strategy employed in the polynomial-time algorithm for the 1-local Hamiltonian problem. $\square$

Assuming Exponential Time Hypothesis for SAT is true, we can prove that there is **no** classical algorithm for problem 4.7 with sub-exponential running time, that is, any algorithm for 3-SAT must take $O^*(2^{\delta n})$ time for $\delta > 0$. The notation O^* indicates the suppression of $\text{poly}(n)$ factor as it will get asymptotically dominated by the exponential function. **Note:** The result implies the worst-case analysis of the problem, or one can create an instance for which any polynomial-time algorithm is unlikely to succeed (assuming ETH is true.)

Theorem 4.8. *If the Exponential Time Hypothesis is true, then there is no polynomial time algorithm for $\{ZZ, Z\}$ -HAMILTONIAN on n -qubits for $m = O(\log^p n)$ with $p > 1$.*

Proof : The proof strategy is to show any algorithm for SAT that saturates the upper bound prescribed by ETH, then the same algorithm can only yield a superpolynomial algorithm for the case when $m = O(\log^p n)$, for any $p > 1$. We evoke the known $\text{poly}(n)$ -time and $O(n)$ -space reduction of 3SAT to MAX2SAT. Also, there is $\text{poly}(n)$ -time and $O(n)$ -space reduction of MAX2SAT to $\{ZZ, Z\}$ -HAMILTONIAN (4.3) problem.

As per ETH, 3SAT over n -variables would require at most $O^*(\alpha^n)$ computational steps (for $\alpha \in (1, 2]$). When the number of clauses is $m = O(\log^{1+\epsilon} n)$, then in the worst-case scenario, we can have $3m$ variables (3 variables per clause). Thus, it would require at most

$$T(n) = \alpha^{3 \cdot \log^{1+\epsilon} n} \quad (31)$$

⁴This brute force algorithm can be enhanced by using a randomized algorithm like the method similar to Schroning's SAT algorithm. This will result in a polynomial speed-up over the brute-force method. In practical cases, this polynomial speed-up might be helpful.

computational steps. Thus, for any $\epsilon > 0$, it would require a super-polynomial number of steps, in the worst case. Since the reduction of 3SAT to $\{ZZ, Z\}$ -HAMILTONIAN causes linear blow-up in terms of the number of variables, the above derived time bound for restricted 3SAT is translated to restricted $\{ZZ, Z\}$ -HAMILTONIAN too. $\square$

4.4.1 Quantumly tractable case

The above analysis has a quantum counterpart as follows.

Problem 4.4 ($\{XX, YY, ZZ, X, Y, Z\}$ -Hamiltonians). *Input: A two local Hamiltonian on n -qubit system which takes the following form*

$$H = \sum_{i,j \in [m]} \alpha_{i,j} X_i \otimes X_j + \beta_{i,j} Y_i \otimes Y_j + \gamma_{i,j} Z_i \otimes Z_j + \sum_{k \in [n]} a_k X_k + b_k Y_k + c_k Z_k \quad (32)$$

Promise : Either, (i) $\lambda_0(H) \leq a$, or (ii) $\lambda_0(H) \geq b$. Along with the promise gap of $b - a \geq 1/\text{poly}(n)$.

The problem is QMA-complete if the parameter $m = \text{poly}(n)$ [CM16]. We prove the problem is in BQP if we restrict the parameter $m = O(\log(n))$.

Theorem 4.9. *The problem 4.4 is in BQP if parameter $m = O(\log(n))$.*

Proof : The complexity of the problem is due to $X_i \otimes X_j + Y_i \otimes Y_j + Z_i \otimes Z_j$ terms in the Hamiltonian. We (informally) call it quantum complexity generating terms. As in the classical case, the 1-local terms do not increase the complexity. There is a subtle difference between the proof for this case and the classical counterpart. Our algorithm is inspired by the proof of $\text{QMA}_{\log} = \text{BQP}$ [MW05]. The idea is to run the Quantum phase estimation algorithm on a maximally mixed state over $m = O(\log(n))$ qubits. The qubits on which only a 1-local Hamiltonian acts can be managed using the strategy employed in the polynomial-time algorithm for the 1-local Hamiltonian problem. $\square$

Theorems 4.7 and 4.9 equip us with a critical characterization, in terms of Pauli string Hamiltonian, for terms whose appearance till a specific frequency makes the input instance tractable. This is sufficient to bring the notion of Fourier analysis, over bounded Hermitian operators, to be an essential tool to design a spectrally-inspired GENERATE GS algorithm.

4.4.2 What if parameter $m > O(\log(n))$?

Assuming some hardness assumption like $\text{BQP} \neq \text{QCMA}$ and Exponential Time Hypothesis for SAT and k -LH, our ground state algorithm can be proved to be optimal in the worst case scenario for $m = O(\log^p(n))$ where $p > 1$. Yet, we have identified some cases in this regime where a polynomial-time algorithm can be provided under specific criteria. We list them as follows. (To be updated)

5 A $\text{poly}(\gamma_i)$ algorithm for GLH_i

5.1 The algorithm

We introduced this parametrized version of the GLH_i problem in the section ???. Since the problem lies in QMAH_i , it is unlikely to have a polynomial-time quantum algorithm. At

the same time, it is unlikely to be as difficult as QMAH_i -complete⁵ problem because there is some evidence for $\text{QMAH}_i \neq \text{QMA}$. Thus, this intermediate-difficulty problem is likely to admit an algorithm with a running time that is intermediate. In this section, we give a quasi-polynomial algorithm for GLH_i .

5.1.1 Our Algorithm for GLH_i Problem

Theorem 5.1 (GLH_i algorithm). *For every instance of GLH_i problem, there exists a quantum algorithm that takes (i) local Hamiltonian $H = \sum_{j=1}^m H_j$, (ii) two parameters a, b such that $(a - b) \geq 1/\delta$ (ii) β_p -Guiding State $|u\rangle$ with $\gamma = \Omega(1/2^{O(\log^p n)})$ as input and decides if:*

- $\lambda_0(H) \leq a$
- $\lambda_0(H) \geq b$

using a quantum circuit of

1. Depth: $d(n) = \text{poly}\left(m, \frac{1}{\delta}, \frac{1}{\gamma}\right)$, and
2. Size: $s(n) = \text{poly}\left(n, m, \frac{1}{\delta}, \frac{1}{\gamma}\right)$.

Proof: The proof of the theorem is given by an explicit construction of an algorithm, followed by the proof of correctness of the algorithm.

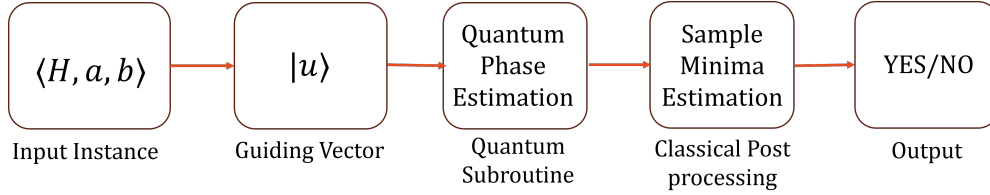


Figure 7: The key steps in universal quantum algorithm design.

- **Step I:** A polynomial depth quantum circuit produces a guided state $|u\rangle$ using the GENERATE GS algorithm.
- **Step II:** The guided state $|u\rangle$ is input into the quantum phase estimation algorithm. We utilize a quantum phase estimation algorithm for this purpose.
- **Step III:** The algorithm is repeated for $\text{poly}\left(\frac{1}{\gamma}\right)$ to collate the sample set S . **Note:** this set contains the minimum eigenvalue as its smallest element (with high success probability).
- **Step IV:** Run the FIND MINIMUM ELEMENT algorithm on the sample set S . Compare the output value with the input parameters a and b , and declare the result accordingly.

The proof of the algorithm's correctness follows directly from the proof of its correctness given in its steps. The correctness of the quantum phase estimation algorithm is given in

⁵Please note, the best known quantum algorithm for the n -qubit LH problem has exponential running time, i.e., $O(2^{n/2})$.

Section ??). The design and analysis of the GENERATE GS and FIND MINIM ELEMENT algorithms are given in Subsection 5.1.2.

In the following three sections, we present the algorithm for GENERATE GS and provide a proof of its optimality (assuming certain complexity-theoretic assumptions).

5.1.2 Generating the Guiding State: GENERATE GS algorithm

This subsection contains the design of the algorithm.

Input to the algorithm:

An efficient description of a 2-Local Hamiltonian matrix: $H = \sum_{i=1}^{\text{poly}(n)} H_i$.

Steps to the algorithm: This is a four-step process to generate a guiding state for the input Hamiltonian H .

- **Step I:** This step will change the representation of the Hamiltonian operator to the Fourier basis. Technically, it implies expanding the bounded Hermitian operator H into Pauli Normal Form. This change of representation is known to be efficient [CM16].

$$H = \sum_{s \in \{0,1,2,3\}^n} \hat{f}_s \chi_s \quad (33)$$

- **Step II:** This step will check the locality of the Pauli Hamiltonian as given in the diagram. If it is 1-Local, then it outputs the ground state using 4.2. Otherwise, it goes to the next step.
- **Step III:** For terms with Pauli degree at most two, check for the threshold $m \leq O(\log(n))$. If it's true, then run the phase estimation on the maximally mixed state

$$\rho_{\text{gs}} := \left(\frac{1}{\sqrt{2^m}} \sum_{i \in \{0,1\}^m} |i\rangle \langle i| \right) \otimes |u\rangle \langle u| \quad (34)$$

where $|u\rangle$ is a product state over the remaining qubits that does not appear in complexity generator Pauli terms, that is $\{ZZ\}$ or $\{XX + YY + ZZ\}$ Pauli terms. If the threshold $m > O(\log(n))$, the fidelity of the ground state with our estimated ground state $\gamma \leq 1/2^m \leq 1/\text{poly}(n)$ ⁶. Yet there are cases for which the maximally mixed state would give a polynomial-time algorithm for the cases mentioned in 4.4.2.

Output of the algorithm: An estimate of the density matrix corresponding to the ground state of the input Hamiltonian H .

Remark: The pair (H, ρ_{gs}) is input to QPE, the algorithm that gives an estimate of the minimum eigenvalue in a time proportional to the fidelity of the estimated ground state to the actual ground state, that is γ .

⁶The fidelity of $\gamma \geq 1/\text{poly}(n)$ is considered desirable; a good guiding state

5.2 Proof of correctness of the algorithm

5.2.1 Resource analysis for GENERATE GS algorithm

Step I and II test the types of the Pauli terms. Thus, the maximum steps are upper bounded by the number of terms in the Hamiltonian, that is $\text{poly}(n)$, because the testing algorithms do a finite number of checks on each $\text{poly}(n)$ Hamiltonian term. This step is purely a classical pre-processing.

Step III will require preparing maximally mixed state. Which can be efficiently simulated by performing quantum phase estimation on half of an input maximally entangled state (for which the reduced density matrix is maximally mixed) and performing fixed-point amplitude amplification to boost the probability of success coherently. Thus, the time taken by our method to prepare the ground state is upper bounded by $\text{poly}(n)$.

5.2.2 Minimum Energy Estimation: QPE Algorithm

This QPE algorithm is used in the same sense and setting as prescribed in Subsection 4.2.1. The details can be directly borrowed from that section without any loss of generality or any conceptual conflict.

6 Robustness of the Hierarchy

6.1 Oracle Separation of Levels in the Hierarchy

Various levels in the QMAH hierarchy can be related as follows.

Theorem 6.1. $\forall i \in \mathbb{N}, \text{QMAH}_i \subseteq \text{QMAH}_{i+1}$

Proof: This asserts any level in the hierarchy, say QMAH_i , can be simulated efficiently by QMA_{i+1} . Notice $\text{QMAH}_i \subseteq \text{BQTIME}(2^{O(\log^i n)})$ while, $\text{QMAH}_{i+1} \subseteq \text{BQTIME}(2^{O(\log^{i+1} n)})$. Since $\text{BQTIME}(2^{O(\log^i n)}) \subseteq \text{BQTIME}(2^{O(\log^{i+1} n)})$, we have $\text{QMAH}_i \subseteq \text{QMAH}_{i+1}$ as a consequence. $\square$

Conjecture 6.1. $\forall i \in \mathbb{N}, \text{QMAH}_i \subsetneq \text{QMAH}_{i+1}$.

We conjecture there exists some problem in QMAH_{i+1} that is not in QMAH_i . Since an unconditional proof for the conjecture would imply $\text{BQP} \neq \text{QMA}$, we resort to providing an oracular argument for the conjecture.

Theorem 6.2. $\forall i \in \mathbb{N}$, there exists an oracle O such that $(\text{QMAH}_i)^O \neq (\text{QMAH}_{i+1})^O$.

Proof: The oracle separation method is inspired by the technique by [KF80] that has been used to separate $\beta_{k+1}\text{P}$ from $\beta_k\text{P}$ using an oracle O_c . The key difference is that we have modified the oracle with Simon's oracle (say O_q) to form a hybrid oracle (say, O_{qc}). Then we argue the hybrid oracle suffices to prove the existence of a problem that is in $(\text{QMAH}_{i+1})^{O_{qc}}$ but not in $(\text{QMAH}_i)^{O_{qc}}$. $\square$

7 Discussion, Conclusion and Outlook

We have proposed two infinite hierarchies in QMA by quantifying quantum nondeterminism as a tunable computational resource. A variant of Local Hamiltonian has been proven to be

a complete natural problem for the classes in the hierarchies. Furthermore, we consider our work to open exciting directions for further research, some of which we will discuss now.

First, similar to the hierarchy between BQP and QMA, one can think of a hierarchy between NP and QMA. We suspect that another variant of the **guidable** $\text{LH}(\gamma, \delta)$ problem could be a potential candidate for complete problems in the hierarchy if the promise gap value δ is defined as a tunable parameter rather than γ , the guidance value. This is mainly due to the fact that $k\text{-LH}(\gamma, \delta)$ with $\delta = O(1)$ is NP-complete while it is QMA-complete for $\delta = 1/\text{poly}(n)$. It would be interesting to look for the implications of this hierarchical characterization on the quantum PCP conjecture, which asks if promise gap amplification is possible for QMA-complete problems.

Second, our current investigation has fine-grained the complexity of the **Local Hamiltonian** problem and its variants. The **Local Hamiltonian** problem is useful in quantum many-body physics; thus, fine-graining might provide insight into those systems. Recently, a study was conducted to relate the complexity of computing certain types of symmetry in quantum states [RLW24]. It has been established that computing a certain form of symmetry of a quantum state is BQP-complete. Thus, it would be interesting to fine-tune the complexity of these problems within our hierarchy and then develop a physical interpretation of this classification.

We believe that the study we have initiated in this work would provide new physical insights and perspectives on many-body quantum systems under consideration.

8 References

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9 Appendix I: A review of Hamiltonian complexity theory

We will briefly look at some important results from quantum complexity theory, like the definition of QMA and QCMA classes⁷ and some versions of the Local Hamiltonian problem. A rigorous account of these classes can be found in [Gha24]. We briefly recap the quantum Merlin-Arthur protocol, as our definitions of hierarchies within QMA are based on this setup. The notion of P-uniform circuit family will be frequently used to have a rigorous definition of all the quantum complexity classes discussed here.

Definition 9.1 (P-uniform quantum circuit family). *A family of quantum circuits $\{Q_n\}$ is called P-uniform if there exists a polynomial-time deterministic Turing machine M which, given as input 1^n , outputs a classical description of Q_n .*

9.0.1 Quantum Merlin-Arthur Protocol

A QMA verification procedure V is P-uniform quantum circuit family $\{V_x : x \in \{0,1\}^*\}$, together with functions $\mathbf{m} : \mathbb{N} \rightarrow \mathbb{N}$ and $\mathbf{k} : \mathbb{N} \rightarrow \mathbb{N}$. The function $\mathbf{m}$ specifies the length of Merlin's message/witness to Arthur, and it is assumed that each circuit V_x acts on $\mathbf{m}(|x|) + \mathbf{k}(|x|)$ qubits. The second function $\mathbf{k}$ specifies the required ancillary qubits.

Protocol: ($\mathbf{m}$ -qubit QMA verification procedure) Consider the following process for a string $x \in \{0,1\}^*$ and a quantum state $|\psi\rangle$ on m qubits:

1. Run the circuit V_x on the input $|\psi\rangle |0^k\rangle$.
2. Measure the first qubit of the resulting state on a standard basis, interpreting the outcome 1 as **accept** and outcome 0 as **reject**.

The probability associated with two possible outcomes will be denoted accordingly by $\Pr[V_x \text{ accepts } |\psi\rangle]$ and $\Pr[V_x \text{ reject } |\psi\rangle]$. Now, the formal definition for the complexity class QMA is as follows.

Definition 9.2 (QMA complexity class). *A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ is in QMA if and only if there exists a P-uniform quantum circuit family $\{V_n\}$ and a polynomial $p(n)$, where V_n takes as input a string $\mathbf{x} \in \{0,1\}^*$ with $|\mathbf{x}| = n$, and a $p(n)$ -qubit quantum state $|\psi\rangle$ and decides on acceptance or rejection of $\mathbf{x}$ such that*

- if $\mathbf{x} \in A_{\text{yes}} \implies \exists |\psi\rangle \in (\mathbb{C}^2)^{\otimes p(n)}, \Pr[V_n \text{ accepts } |\psi\rangle] \geq 2/3$.
- if $\mathbf{x} \in A_{\text{no}} \implies \forall |\psi\rangle \in (\mathbb{C}^2)^{\otimes p(n)}, \Pr[V_n \text{ accepts } |\psi\rangle] \leq 1/3$.

One can easily see that both the witness and verification algorithms require only a polynomial amount of quantum resources with respect to the input size. Merlin can use a $\text{poly}(n)$ number of qubits to encode the witness, while Arthur can use a polynomial-depth quantum circuit for the verification. The complexity class QCMA deals with the case where Merlin's proof is restricted to be a classical string of $\text{poly}(n)$ size, while Arthur continues to be a $\text{poly}(n)$ depth quantum circuit.

Definition 9.3 (QCMA complexity class). *A promise problem $A = (A_{\text{yes}}, A_{\text{no}})$ is in QCMA if and only if there exists P-uniform quantum circuit family $\{V_n\}$ and a polynomial $p(n)$, where V_n takes as input a string $\mathbf{x} \in \{0,1\}^*$ with $|\mathbf{x}| = n$, and a $p(n)$ -bit string $\mathbf{z} \in \{0,1\}^{p(n)}$ and decides on acceptance or rejection of $\mathbf{x}$ such that*

⁷They are promise-QMA and promise-QCMA, respectively. See [Gha24] to discern them.

- if $\mathbf{x} \in A_{\text{yes}} \implies \exists |\mathbf{z}\rangle \in \{0,1\}^{p(n)}$, $\Pr[V_n \text{ accepts } |\mathbf{z}\rangle] \geq 2/3$
- if $\mathbf{x} \in A_{\text{no}} \implies \forall |\mathbf{z}\rangle \in \{0,1\}^{p(n)}$, $\Pr[V_n \text{ accepts } |\mathbf{z}\rangle] \leq 1/3$.

It is an open problem if QMA and QCMA are indeed distinct classes. The primary difference lies in the nature of the witness. There is some evidence that quantum witnesses should give a richer class [VV16]. In a relativized world, they are distinct because of a quantum oracle that separates QMA from QCMA [AK07].

9.0.2 k -Local Hamiltonian (LH) Problem

We will frequently discuss the local Hamiltonian problem and several of its variants throughout the paper. A formal definition of the (promise) problem is as follows.

Problem 9.1 (k -Local Hamiltonian problem: $k\text{-LH}(\delta)$). *The promise problem k -Local Hamiltonian is defined as a tuple $\langle H, a, b \rangle$.*

Input: *We are given a k -local Hamiltonian on n qubits $H = \sum_{j=1}^r H_j$ with $r = \text{poly}(n)$. Each Hermitian matrix $H_j \in \mathcal{H}_k$ has a bounded operator norm $\|H_j\| \leq 1$, and its entries are specified by $\text{poly}(n)$ bits. In addition, we are given two constants a and b with $|b - a| \geq \delta > 0$.*

Promise: *For any given instance, either of the two cases below holds.*

- For YES instances, $\lambda_0(H) \leq a$.
- For NO instances, $\lambda_0(H) \geq b$.

Output: *Decide which is the case.*

This problem has been shown to be complete in the class QMA when the parameters k and δ lie within a certain range.

Theorem 9.1 ([KKR04]). *$2\text{-LH}(\delta)$ problem with the promise gap $\delta \geq (\frac{1}{\text{poly}(n)})$ is QMA-complete under Karp reduction.*

$k\text{-LH}$ problem is a quantum generalization of Constraint Satisfaction Problems; the classical problem is the special case where each matrix H_j is diagonal in the computational basis and only contains 0's and 1's. The $k\text{-LH}$ problem has several variants with varying difficulty levels. Now, we will see some of its important variants. Later, we will design a couple of new variants of $k\text{-LH}$ to incorporate them into our hierarchies in QMA.

9.0.3 Some Variants of $k\text{-LH}$ Problem

Variant I: Guided k -Local Hamiltonian problem: $\text{GLH}(k, \gamma, \delta)$. Like QMA, there are several complete problems for QCMA. We especially mention a particular QCMA-complete problem named the Guidable local Hamiltonian problem (GaLH). To fully appreciate this problem, we need to consider a precursor problem, the Guided Local Hamiltonian (GLH) problem, which is BQP-complete. This was introduced by Gharibian & Le Gall [GLG22]. They formalized the complexity of $k\text{-LH}(\gamma, \delta)$ with a good guiding state. We give a brief exposition of the GLH problem.

The definition of GLH consists of three important terminologies as follows:

Subset State ([GKS15]): For any subset $S \subseteq [2^n]_1$, we define the subset state $|S\rangle \in \mathbb{R}^{2^n}$, as the uniform superposition over the elements of S . More precisely,

$$|S\rangle := \frac{1}{\sqrt{|S|}} \sum_{i \in S} |i\rangle. \quad (35)$$

Guided State: For a given k -LH problem with Hamiltonian H , the unit norm vector $|\mathbf{u}\rangle \in \mathbb{R}^{2^n}$ is defined to be a guided state if it is a subset state

$$|\mathbf{u}\rangle := \frac{1}{\sqrt{|S|}} \sum_{i \in S} |i\rangle$$

with a non-empty subset $S \subseteq [2^n]_1$ and $|S| = O(\text{poly}(n))$. The last restriction on the size of the subset S is crucial to discern a subset state from a guided state.

Guidance Value: For a given k -LH problem with the Hamiltonian H , the scalar $\gamma(n) \in [0, 1]$ is called the guidance value of the guided state $|\mathbf{u}\rangle$

$$\gamma(n) := \|\Pi_{\text{gs}} |\mathbf{u}\rangle\|^2$$

where Π_{gs} is the projection on the subspace spanned by the ground state of H . This quantifies the quality of a guided state. Having these three terminologies at hand, the problem is defined as follows.

Problem 9.2 ([GLG22]: GLH(k, γ, δ) problem). *The problem is a tuple $\langle H; |\mathbf{u}\rangle; (a, b) \rangle$ with the following characteristics.*

Input: (i) a k -local Hamiltonian matrix $H = \sum_{j=1}^m H_j$ for an n -qubit system such that $\|H\| \leq 1$, (ii) an efficient description of a semi-classical state $\mathbf{u} \in \mathbb{C}^{2^n}$ and (iii) promise gap parameter $a, b \in \mathbb{R}$ such that $|b - a| \geq \delta > 0$.

Promise:

- Fidelity with the ground space $\|\Pi_{\text{gs}} |\mathbf{u}\rangle\| \geq \gamma$.
- Either case (i) $\lambda_0 \leq a$ or (ii) $\lambda_0 \geq b$ holds.

Output:

1. If $\lambda_0(H) \leq a$, output YES.
2. If $\lambda_0(H) \geq b$, output NO.

Within a certain parameter range, this problem is known to be BQP-complete. Currently, the best-known result is summarized in the following theorem.

Theorem 9.2 ([CFW22]). : For any $\gamma(n) \in (0, 1 - \Omega(1/\text{poly}(n)))$, constant $k \geq 2$, there exists a promise gap parameter $a, b \in [-1, 1]$ with $b - a \in \Omega(1/\text{poly}(n))$ such that GLH(k, γ, δ) is BQP-complete.

Variant III: Guidable k -Local Hamiltonian problem: GaLH(k, γ, δ). The idea of Guidable Local Hamiltonian (GaLH) was introduced by Weggemans et al. [WFC23]. The key difference between GLH and GaLH is that GLH has the guiding state given as part of the problem. But in GaLH, there is a mere promise of the existence of such a guiding state.

Problem 9.3 (Guidable k -Local Hamiltonian (GaLH) problem). *The problem is a tuple $\langle H; (a, b) \rangle$ with the following characteristics.*

Input: *The inputs given are (i) k -local Hamiltonian $H = \sum_{j=1}^m H_j$ acting on n qubits such that $\|H\| \leq 1$, (ii) a fidelity parameter $\gamma(p) \in (0, 1]$, and (iii) a promise gap parameter $a, b \in \mathbb{R}$ such that $|b - a| \geq \delta > 0$.*

Promise: *We have that either $\lambda_0(H) \leq a$ or $\lambda_0(H) \geq b$ holds, where $\lambda_0(H)$ denotes the ground state energy of H .*

Extra promise : *Denote Π_{gs} for the projection on the subspace spanned by the ground state of H . Then, there exists a unitary V implemented by a quantum circuit composed of at most $T = O(\text{poly}(n))$ gates from a set of fixed gates $\mathcal{G}$ that produces the state $|\phi\rangle = V|0\rangle$, which has $\|\Pi_{gs}\mathbf{u}\| \geq \gamma$.*

Output:

1. *If $\lambda_0(H) \leq a$, output YES.*
2. *If $\lambda_0(H) \geq b$, output NO.*

Theorem 9.3 ([WFC23]). *For all $p \in \mathbb{N}$, $k \geq 2$, $\gamma \in (\frac{1}{\text{poly}(n)}, 1 - \frac{1}{\text{poly}(n)})$, and $\delta = \Omega(1/\text{poly}(n))$, the GaLH(k, γ, δ) problem is a complete problem for QCMA under randomized reduction.*

This result has some very significant implications. Since P, BQP, and NP are contained in the QCMA, an efficient description of the guiding state must exist if the problem of these classes is reduced to a local Hamiltonian problem. Thus, the bottleneck to solving these problems reduces to finding a guiding state⁸. Once we have the guiding state, there is an efficient quantum algorithm to compute the minimum eigenvalue of the problem. See Section 5.1 for further analysis.

10 Appendix II: A review of an infinite hierarchy between P and NP

Ladner proposed a hierarchy between P and NP assuming $P \neq NP$ [Lad75] is true. But the problems in the hierarchy were mathematically contrived. Kintala & Fischer [KF80] defined another hierarchy using the concept of a Turing machine with access to a limited amount of classical nondeterminism. Later, the hierarchy was populated with several natural problems, such as a specific variant of SAT, HAM-CYCLE, and VERTEX-COVER [Far94].

10.0.1 Classical β -Hierarchy

Informally, β -P hierarchy is defined as classes of sets accepted by a polynomial-time bounded Turing machine that makes a poly-logarithmic number of non-deterministic steps. By varying the degree of the poly-logarithmic function, we get an infinite hierarchy between P and NP-complete sets. The next two definitions formalize this notion.

Definition 10.1 (β -Hierarchies [KF80]). *Let $\mathcal{F}$ be a class of functions, and $\mathcal{C} = \bigcup_{f \in \mathcal{F}} \text{DTIME}(f)$ be the class of sets accepted by deterministic $\mathcal{F}$ -time bounded Turing machines. For the total function $\mathbf{g} : \mathbb{N} \rightarrow \mathbb{N}$, $\mathbf{g}(n)\text{-}\mathcal{C}$ denotes the class of sets accepted by nondeterministic $\mathcal{F}$ -time bounded Turing machines making at most $\mathbf{g}(n)$ nondeterministic steps on every input of length n .*

⁸This means the complexity of solving the problems in QCMA can be squeezed to finding a 'good' guiding state for the reduced Local Hamiltonian problem.

Definition 10.2 (β -P Hierarchy [KF80]). *For every $k \in \mathbb{N}$, by definition the complexity class $\beta_k P = \bigcup_{c \in \mathbb{N}} (c \cdot \log^k n) \cdot P$. Then βP -hierarchy is $\beta P = \bigcup_{k \in \mathbb{N}} \beta_k P$.*

As a consequence of the above definition, the complexity class P can be thought of as a class of problems that a polynomial-time Turing machine can solve with $O(\log n)$ nondeterministic steps.

Theorem 10.1 ([KF80]). *With respect to Kintala's βP hierarchy, $P = \beta_1 P$.*

Proof: The proof follows from the fact that the Turing machine can brute force through the space of $O(\log n)$ non-deterministic guesses. **Note:** there are at most $2^{O(\log n)} \leq \text{poly}(n)$ possibilities to be explored by the Turing machine. $\square$

Subsequent studies have identified several natural problems at various levels of the hierarchy. For example, G. Farr proposed a particular variant of the SAT problem and proved it to be complete for the various levels.

Problem 10.1 (β_k -SAT [Far94]). *It is defined as the set of all pairs $\langle \phi, V \rangle$ of the formula ϕ and the set V of at most $\log^k |\phi|$ variables in ϕ , for which there exists an assignment $\mathcal{A}$ to all variables in V that forces a satisfying assignment for ϕ . Note that β_1 -SAT is p -isomorphic to the P -complete complement of the unit resolution problem.*

Theorem 10.2 ([Far94]). *For every $k \in \mathbb{N}$, β_k -SAT is a complete problem for $\beta_k P$ under polynomial-time many-one reduction.*